Lab 2: A neural network from scratch (Ch. 3)

Reading and Learning Outcomes

Reading: Reddi, Chapter 3 DL Primer: §3.4 Neural Network Fundamentals; §§3.5.1–3.5.5 (training overview, forward propagation, loss functions, backward propagation, optimization); and §3.6 Prediction Phase.

After completing this lab, students should be able to:

- Design a two-hidden-layer neural network with dimensionally correct parameters.
- Implement forward propagation, cross-entropy loss, and backpropagation from scratch.
- Validate analytical gradients using numerical gradient checking.
- Compare learning rates, hidden widths, accuracy, and forward/backward timing.

1. [10] Dataset: `load_digits`. Scale inputs by dividing by 16. One-hot encode labels. Split 70/30 stratified with `random_state=42`.
2. [15] On paper, design the network. Use two hidden layers. Input 64 → hidden width W → ReLU → hidden width W → ReLU → output 10 → softmax. Use W = 16 and W = 64. Weight shapes: `(64,W)`, `(W,W)`, `(W,10)`. Bias shapes: `(W,)`, `(W,)`, `(10,)`. Initialise weights from `N(0, 0.01)`.
3. [20] Implement the forward pass layer by layer. Check output shape is `(batch, 10)` and each row sums to 1 within `1e-6`.
4. [10] Implement cross-entropy loss. Sanity check: untrained network loss should be about `ln(10) = 2.3026`.
5. [30] Derive gradients by hand with the chain rule. Implement them. Verify with numerical gradient checking: nudge 10 random weights by `±1e-5`, compare analytic and numerical slopes. Relative error must be `<1e-6` for at least 9 of 10 weights.
6. [20] Write the training loop: shuffle, mini-batch size 32, 50 epochs, per-epoch logging. Run with learning rates `[0.001, 0.01, 0.1]` and widths `[16, 64]`. Plot loss curves to `results/loss_curves.png`. Record final test accuracy.
7. [10] Time forward and backward passes separately. Median of 100 runs. Explain which dominates and link it to matrix multiplication.

Graded evidence: passing gradient check, loss curves, timing breakdown.